

Too Late to Train, Too Early To Use?

A Study on Necessity and Viability of Low-Resource Bengali LLMs

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Abstract

Each new generation of English-oriented Large Language Models (LLMs) exhibits enhanced cross-lingual transfer capabilities and significantly outperforms older LLMs on low-resource languages. This prompts the question: *Is there a need for LLMs dedicated to a particular low-resource language?* We aim to explore this question for Bengali, a low-to-moderate resource Indo-Aryan language native to the Bengal region of South Asia.

We compare the performance of open-weight and closed-source LLMs such as LLaMA-3 and GPT-4 against fine-tuned encoder-decoder models across a diverse set of Bengali downstream tasks, including translation, summarization, paraphrasing, question-answering, and natural language inference. Our findings reveal that while LLMs generally excel in reasoning tasks, their performance in tasks requiring Bengali script generation is inconsistent. Key challenges include inefficient tokenization of Bengali script by existing LLMs, leading to increased computational costs and potential performance degradation. Additionally, we highlight biases in machine-translated datasets commonly used for Bengali NLP tasks. We conclude that there is a significant need for a Bengali-oriented LLM, but the field currently lacks the high-quality pretraining and instruction-tuning datasets necessary to develop a highly effective model.*

1 Introduction

The release of GPT-3.5 (Brown et al., 2020) in late 2022 has kickstarted the current era of rapid progress in Large Language Models (LLMs). However, this progress is not merely a result of increased model scale, rather, it stems from a virtuous

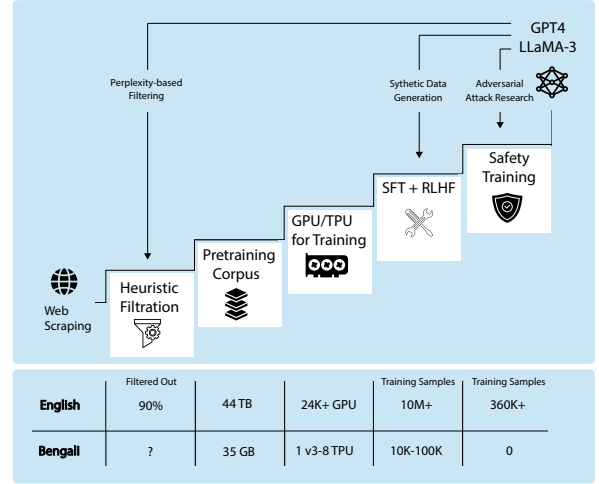


Figure 1: Large Language Model Training Pipeline and Resource Comparison between BanglaT5 (Bhattacharjee et al., 2022) vs. LLaMA-3 (Meta, 2024). The first step towards capable Bengali LLMs is collecting a large pretraining corpus. However, the iterative nature of LLM development makes it unlikely that sufficient pretraining data and compute alone would enable Bengali LLMs to match the capabilities of their English-oriented counterparts.

cycle of innovation, where lessons from each generation inform the development of the next. Techniques such as synthetic data generation (Eldan and Li, 2023; Gunasekar et al., 2023), the integration of mathematical and coding tasks to enhance reasoning capabilities (Ma et al., 2023), and research into adversarial attacks (Zou et al., 2023) for improved safety have all contributed to the ever-increasing capabilities of LLMs. As illustrated in Figure 1, developing state-of-the-art LLMs involves filtering vast amounts of web-scraped data, utilizing substantial computational resources, and implementing advanced techniques for alignment and safety.

However, this progress poses a dilemma for low-resource languages like Bengali. Despite being one of the most widely spoken languages, the size of Bengali pretraining and instruction-tuning data are

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* The evaluation code is available at <https://github.com/satak100/BanglaBench>.

minuscule compared to their English counterparts (Hasan et al., 2020; Bhattacharjee et al., 2021). To this date, BanglaT5 (Bhattacharjee et al., 2022), a 248 million parameter encoder-decoder T5 transformer (Raffel et al., 2020), remains the most capable Bengali Language Model. Furthermore, prematurely investing in training larger models might yield lackluster results due to the lack of high-quality Bengali data.

In this study, we aim to quantify the demand and viability of a Bengali-oriented LLM. To this end, we compile a representative benchmark of both Natural Language Understanding (NLU) and Natural Language Generation (NLG) downstream tasks for Bengali and evaluate a wide range of open-weights and closed-source models. Our key findings include:

1. Compared to fine-tuned BanglaT5 or BanglaBERT, English-oriented LLMs excel in comprehension tasks (NLU) and perform inconsistently in Bengali generation (NLG).
2. Using machine translation to translate English NLG datasets into Bengali biases the dataset towards specific writing styles and skews downstream metrics such as BLEU and ROUGE in favor of fine-tuned models regardless of generation quality.
3. Bengali is over-tokenized by the BPE tokenizer used English LLM, with an average of ~ 0.85 characters-per-token compared to ~ 4.5 for English. Over-tokenization leads to $O(n^2)$ attention-based LLMs being highly inefficient in processing Bengali script.
4. The outputs of English LLMs on Bengali Reward Modeling tasks do not correlate strongly with human judgment. As such, these LLMs have limited applicability in generating Bengali RLHF datasets.

A comprehensive evaluation of state-of-the-art LLMs on 7 Bengali NLU and NLG tasks, revealing task-dependent performance variations. An analysis of the inefficient tokenization of Bengali script by existing LLMs and its impact on model performance. Insights into the challenges and potential strategies for developing Bengali-specific LLMs, balancing the need for language-specific models against the rapid progress in multilingual capabilities of existing LLMs.

2 Preliminaries

2.1 Tasks and Datasets

We evaluate the latest LLMs on a wide range of Bengali downstream tasks as shown in Table 1, covering both Natural Language Understanding (NLU) and Natural Language Generation (NLG). We elaborate on the differences between the Question-Answering datasets and the construction of the Reward Modeling dataset.

Question-Answering: Among the Question-Answering (QA) datasets, Squad-bn (Bhattacharjee et al., 2021) and BanglaRQA (Ekram et al., 2022) are close-ended reading comprehension datasets, i.e. the LLM is given a context and a question, and must first determine whether the answer is present in the context and then extract the answer if it does. BENQA is a close-ended, open-domain QA dataset where the LLM is asked a factual STEM-related question from the middle-school/high-school curriculum of Bangladesh.

Reward Modeling: While combined fine-tuning downstream tasks (Chung et al., 2024) such as translation, summarization, and Question-Answering was the dominant post-pretraining paradigm from early Language Models such as T5 (Raffel et al., 2020), much of the impressive capabilities of Billion parameter-scale LLMs can be attributed to RLHF (Ouyang et al., 2022a,b), which improves the generalizability of LLMs even to unseen tasks. Lee et al. (2023) has shown that feedback from other LLMs can substitute the need for human feedback in RLHF, in a method dubbed RLAIIF. RLAIIF can also be more robust than simple synthetic fine-tuning data generation (Abdin et al., 2024) which might overfit benchmarks (Zhang et al., 2024). To test the capability of English LLMs to provide feedback on Bengali NLG, we created a new Reward Modeling task based on XLSum (Hasan et al., 2021a). In this abstractive summarization dataset, we give the LLM a Bengali article and two summaries and ask it to pick the better one. We take the summary in the XLSum as the gold summary and the article’s first sentence as the heuristically best summary. We instruct the LLM to prefer abstractive summaries over extractive ones. Refer to Appendix B for the instruction template used. We randomly pick 300 samples from the test dataset due to cost considerations.

Type	Task	Dataset	Test	Data Curation	Metric	Best Model
NLG	Translation	BanglaNMT (Hasan et al., 2020)	1000	aligned	BLEU	LLaMA-3-70B (B-E) NLLB-3.3B (E-B)
	Monolingual Summarization	XLSum (Hasan et al., 2021a)	1012	in-language	ROUGE-2	BanglaT5-248M-FT
	Crosslingual Summarization	CrossSum (Hasan et al., 2021b)	161 (E-B) 161 (B-E)	aligned	ROUGE-2	LLaMA-3-70B (E-B) LLaMA-3-70B (B-E)
	Paraphrase	BanglaParaphrase (Akil et al., 2022)	23332	machine translated	ROUGE-2	BanglaT5-248M-FT
NLU	QA (compr.)	Squad-bn/BQA (Bhattacharjee et al., 2021)	2504	machine translated	F1/Match	LLaMA-3-70B
	QA (compr.)	BanglaRQA (Ekram et al., 2022)	1493	in-language	F1/Match	LLaMA-3-8B-q4-FT
	QA (open-dom.)	BEnQA (Shafayat et al., 2024)	5161	in-language	Acc.	GPT4
	Inference	XNLI-bn (Bhattacharjee et al., 2021)	4895	machine translated	Acc.	LLaMA-3-8B-q4-FT
	Reward Modeling	XLSum (adapted subset)	300	in-language	Acc.	LLaMA-3-70B

Table 1: Bengali datasets used in our experiments and the best model for each dataset. E-B stands for *English-to-Bengali* generative tasks. FT stands for *finetuned*.

2.2 Models

Large Language Models can be categorized into open weights or closed-source models, based on whether individual users can download the model parameters or not. The current state-of-the-art LLM according to most benchmarks and user preference (Chiang et al., 2024) is the closed-source GPT4o. The leading open-weights LLM in LLaMA-3-70B-Instruct (Meta, 2024), which ranks 9th on the English-only [LMSYS Leaderboard](#) and 12th overall. We note that open-weights models are not *open-source* because most have proprietary licenses that restrict certain use cases such as commercial applications or synthetic data generation.

Open-weights Models: We test both the 8 and 70 billion variants of LLaMA-3 on all downstream tasks. We selectively include results from other open-weights LLMs such as Mistral-7B-0.3 (Jiang et al., 2023), Aya-23-8B (Aryabumi et al., 2024), Qwen-2-72B (Bai et al., 2023) for certain tasks. Aya-23 is a multilingual LLM family that was *not* specifically trained for Bengali but was trained on related language families. Qwen-2 is a primarily English-Chinese LLM family with Bengali-specific training data augmentation [†].

[†]<https://qwenlm.github.io/blog/qwen2/>

For translation, we test 3 variants of the translation-only NLLB Language Model (Costa-jussà et al., 2022) on BanglaNMT (Hasan et al., 2020). We also test the performance of 8-bit quantized LLaMA-3-8B-Instruct to showcase what is possible on consumer-grade hardware. **All reported models are Chat- or Instruct-tuned unless specified otherwise.**

Closed-Source Models: Due to high inference costs on Bengali text, we report GPT4o performance only on the Reward Modeling task. We report the performance of closed-source models such as GPT3.5, GPT4, and Gemini-1.5-Pro (Team et al., 2023) if present in the literature.

2.3 Tokenization of Bengali Script

Almost all LLMs use some variant of Byte-Pair Encoding (BPE) (Sennrich et al., 2015), an algorithm that iteratively combines the most common substrings into tokens. This naturally leads to under-represented scripts and notations being tokenized at higher granularities, leading to undertrained tokens, lower efficiency, and information density.

We run pilot experiments on how LLMs tokenize Bengali text using the articles in XLSum (Hasan et al., 2021a). We find that the average character-per-token value for Bengali using

English LLMs is ~ 0.85 , which means that each token corresponds to *less* than one Unicode Bengali character. In comparison, the character-per-token value for English is ~ 4.5 . The notable exception is BanglaT5 (Bhattacharjee et al., 2022) which trained the tokenizer mainly on Bengali text and NLLB (Costa-jussà et al., 2022), which upsampled low-resource languages and downsampled high-resource ones when training the tokenizer. Detailed findings are presented in Appendix C.

Doddapaneni et al. (2022) notes the excessive tokenization of Bengali by BERT-based models. Yuan et al. (2023) highlight a novel link between excessive tokenization and the subpar performance of finetuning on languages that use a non-Latin script. They further highlight the existence of redundant tokens and show that removing them improves finetuning results.

3 Experimental Setup

We use Together AI API for full-precision inference with open-weights LLMs. For NLLB and 8-bit quantized LLaMA-3-8B, we use Hugging Face library on a single NVIDIA RTX A6000 machine. For Aya-23-8B, we use a $3\times$ NVIDIA RTX A6000 cluster. We use the sacreBLEU library to calculate BLEU and the Multilingual-rouge-scoring repository for ROUGE scores on Bengali text. For summarization tasks (Hasan et al., 2021a,b), we truncate the articles to 7000 tokens using the LLaMA-3 tokenizer due to the 8192 context window of LLaMA-3 models.

In tasks where frozen LLMs underperform, we minimally fine-tune LLaMA-3-8B-Instruct to probe the limitations of LLMs. We finetune LLaMA-3-8B-Instruct using 4-bit integer quantization, QLoRA (Dettmers et al., 2024) using the Unsloth AI library on a single NVIDIA RTX A6000. Task-wise hyperparameters are in Appendix A.

4 Results

In this section, we cover the results of our experiments on NLG and NLU tasks sequentially. The excellent performance of the NLLB-1.3B-Distilled (Costa-jussà et al., 2022) on Bengali-to-English translation as highlighted in Section 4.1 and the lackluster performance of even GPT4o on Reward Modeling in Section 4.6 are particularly noteworthy since both are relevant to synthetic data generation and RLHF required to train LLMs.

4.1 Translation

Table 2 shows that Google Translate significantly outperforms all LLMs and encoder-decoder transformers on both Bengali-to-English (B-E) and English-to-Bengali (E-B) translations. The large difference between Google Translate and other LLMs potentially points to data contamination. On the FLORES-101 (Goyal et al., 2022) Bengali dev. test, NLLB-200 models were significantly better than Google Translate (Costa-jussà et al., 2022).

LLaMA-3-70B is the most capable B-E translator among the open-weights models, beating out the finetuned BanglaT5 (Bhattacharjee et al., 2022). This result disagrees with Asai et al. (2023) where small, finetuned encoder-decoder models outperformed LLaMA-2 on other tasks. Perhaps more impressively, the translation-specialized NLLB-3.3B (Costa-jussà et al., 2022) is the best E-B translator, with even smaller NLLB variants outperforming much larger LLMs. As highlighted in Appendix Table 8, The NLLB model family also boasts better tokenization support for Bengali, further improving inference speed and efficiency. Notably, the largest NLLB model is NLLB-54B-MoE which performs even better. See Appendix Table 10 for the comparison of NLLB variants on another English-Bengali dataset.

The consistent difference between E-B and B-E underlines how all translation systems find it harder to generate Bengali script (E-B) than to understand it (B-E).

Model	B-E	E-B
BanglaT5-248M-FT	31.30	17.40
NLLB-600M-dis.	29.52	17.56
NLLB-1.3B-dis.	30.96	18.97
NLLB-3.3B	30.97	19.73
Mistral-7B-v0.3	14.91	3.67
LLaMA-3-8B-q8	26.82	12.07
LLaMA-3-8B	28.48	12.82
LLaMA-3-70B	33.55	18.92
Qwen-2-72B	32.68	14.34
Google Translate	38.58†	28.15†

Table 2: Bengali-to-English (B-E) and English-to-Bengali (E-B) Translation performance of different models on BanglaNMT (Hasan et al., 2020). Reporting BLEU scores. † Google Translate API was used on June 21, 2024. The large BLEU score gap suggests data contamination in the Google Translate engine.

4.2 Summarization

In Table 3, we show that the finetuned BanglaT5 (Bhattacharjee et al., 2022), a 248M encoder-decoder performs better than even LLaMA-3-70B, a $320\times$ larger English LLM on Bengali-to-Bengali (B-B) summarization. B-B summarization requires both Bengali reading comprehension and generation. On the other hand, LLaMA-3-70B has $2\times$ higher ROUGE-2 score than BanglaT5 on B-E cross-lingual summarization and outperforms it on E-B summarization as well. Even the smaller 8B LLaMA-3 variant outperforms BanglaT5 on B-E CrossSum while Qwen-2-72B performs on par with the similarly sized LLaMA-3.

Dataset	Model	B-B	B-E	E-B
XLSum	BanglaT5-FT	13.7	-	-
	Mistral-7B-v0.3	6.40	-	-
	LLaMA-3-8B	7.36	-	-
	LLaMA-3-70B	8.66	-	-
	Qwen-2-72B	7.54	-	-
CrossSum	BanglaT5-FT	-	6.40	4.00
	Mistral-7B-v0.3	-	5.61	3.21
	LLaMA-3-8B	-	8.88	2.75
	LLaMA-3-70B	-	12.83	4.93
	Qwen-2-72B	-	12.54	4.91

Table 3: ROUGE-2 scores of LLMs on XLSum (Hasan et al., 2021a) and CrossSum (Hasan et al., 2021b). B-B denotes Bengali Article-to-Bengali summaries.

4.3 Paraphrasing

Dataset	Model	BLEU
Bangla-Paraphrase	BanglaT5-FT	32.80
	LLaMA-3-8B-q8	8.21
	LLaMA-3-8B-q4-FT	26.99
	LLaMA-3-8B	9.13
	LLaMA-3-70B	10.18
	Qwen-2-72B	12.47

Table 4: Performance of different models on BanglaParaphrase (Akil et al., 2022).

Table 4 shows the finetuned BanglaT5 (Bhattacharjee et al., 2022) outperforms all LLMs on Bengali paraphrase generation. As with B-B summarization, BanglaParaphrase (Akil et al., 2022) is also a Bengali-to-Bengali task. However, BanglaT5’s BLEU metric is $3\times$ higher than even the LLaMA-3-70B. We manually inspected the

reference paraphrase in the dataset and BanglaT5’s and LLaMA-3-70B outputs. We discovered that the paraphrases generated by BanglaT5’s outputs were more similar to the reference paraphrase in word choice, succinctness, and grammatical structure, while LLaMA-3-70B generated different but still perfectly valid paraphrases, with a slight tendency to generate longer phrases. Therefore, we suspect the high BLEU score of BanglaT5 to the fact that BanglaParaphrase was generated synthetically using translation and back-translation. Specifically, Akil et al. (2022) used the translation model introduced by Hasan et al. (2020) to generate 5 paraphrases of each Bengali sentence in their corpus and filtered using LaBSE (Feng et al., 2022). Both the translation pipeline and the choice of filtration likely introduce grammatical and word-choice biases into the dataset.

To investigate our suspicion, we run a small-scale fine-tuning experiment on LLaMA-3-8B-Instruct. We finetune LLaMa-3 using 4-bit quantization and QLoRA (Dettmers et al., 2024) for only 1 epoch on the 420K training samples from BanglaParaphrase. [‡] Despite using int-4 quantization and QLoRA, our fine-tuned LLaMA-3-8B-q4-FT significantly outperformed all non-finetuned LLMs including LLaMA-3-70B. Through manual inspection, we find that LLaMA-3-8B-q4-FT generates phrases similar to the reference paraphrase, with overlapping word choice and grammatical structure. Therefore, we surmise that the use of machine translation and LaBSE filtering has biased the reference summaries in BanglaParaphrase towards a certain linguistic style. As such, we advocate for human evaluation (Stiennon et al., 2020) over automated metrics such as BLEU or ROUGE for synthetic NLG tasks.

4.4 Question-Answering

Out of the 3 QA datasets tested, Squad-Bn (Bhattacharjee et al., 2021) and BanglaRQA (Ekram et al., 2022) are reading comprehension tasks where a passage is provided and the models must answer with a single substring/span of the passage. Squad-bn and BanglaRQA have non-answerable questions, i.e. the answer is not in the passage. Furthermore, BanglaRQA contains questions where the answers are yes-no or multiple spans from the passage.

[‡]In contrast, BanglaT5 was fine-tuned for 10 epochs on 551K masking-augmented training samples in full-precision.

Dataset	Model	F1	Exact
Squad-Bn	BanglaT5-FT	74.8	68.5
	Mistral-7B-v0.3	54.9	49.8
	LLaMA-3-8B-q8	75	68.5
	LLaMA-3-8B	75.5	68.8
	LLaMA-3-70B	81.9	75.8
	Aya-23-8B	36.8	29.4
BanglaRQA	BanglaBERT-FT	63.2	47.6
	BanglaT5-FT	78.1	62.4
	LLaMA-3-8B	69.2	52.7
	LLaMA-3-8B-q4-FT	80	65.8
	LLaMA-3-70B	72.2	52.1
BEnQA	LLaMA-3-8B	-	45.7
	LLaMA-3-70B	-	64.8
	GPT3.5†	-	37.2
	GPT4†	-	75.1

Table 5: Bengali Question-Answering performance of LMs on Squad-bn (Bhattacharjee et al., 2021), BanglaRQA (Ekram et al., 2022) and BEnQA (Shafayat et al., 2024). Reporting accuracy in the “Exact” column for BEnQA. † Results from Shafayat et al. (2024).

We instruct LLMs to determine if the answer exists in the context passage instead of answering directly from their parametric memory. For BanglaRQA, we instruct the LLM to determine the type of answer it should produce (yes-no, single-span, or multi-span) before writing the actual answer. See Appendix B for the exact prompts used. Table 5 shows that both LLaMA variants outperform the fine-tuned finetuned BanglaT5 on Squad-Bn. LLaMA-3-70B, in particular, shows convincing improvements in F1 (+7.1) and Exact Match (+7.3) metrics. However, in BanglaRQA, the fine-tuned BanglaT5 outperformed non-finetuned LLM by large margins. We manually inspected the LLaMA-3-70B’s output and found it was prone to misclassifying yes-no and multiple-span questions as single-span questions.

We fine-tuned LLaMA-3-8B-Instruct using QLoRA and 4-bit integer quantization for 3 epochs on the BanglaRQA train set. LLaMA-3-8B-q4-FT outperformed fine-tuned BanglaT5 by 1.9 units higher F1 and 3.4 percent higher Exact Matches.

BEnQA (Shafayat et al., 2024) is an open-domain, multiple-choice QA dataset collected from the high school STEM curriculum of Bangladesh. Table 5 shows that GPT4 leads LLaMA-3-70B by a significant margin (+10.3). Notably, the much smaller LLaMA-3-8B outperforms GPT3.5 by 8.5 points, despite GPT3.5 and GPT4 using the same

tokenizer. This suggests that the effect of over-tokenization 2.3 is less pronounced on NLU tasks.

4.5 Natural Language Inference

Dataset	Model	Acc.
XNLI-bn	BanglaBERT-FT	82.8
	Mistral-7B-v0.3	47.4
	LLaMA-3-8B-q8	54.9
	LLaMA-3-8B-q4-FT	83.1
	LLaMA-3-8B	57.3
	LLaMA-3-70B	64.6
	Qwen-2-72B	61
XNLI-bn † (300 subset, 15-shot)	GPT-3.5 Turbo	92
	Gemini 1.5 Pro	91.5

Table 6: Bengali Natural Language Inference performance of LMs. † results from Faria et al. (2024).

Table 6 shows a significant gap between finetuned and non-finetuned models on Natural Language Inference. Due to the large gap in performance between the finetuned BanglaBERT-111M (Bhattacharjee et al., 2021) and LLaMA-3-70B, we minimally finetuned LLaMA-3-8B using parameter-efficient methods to probe the reason. Our finetuned LLaMA-3-8B-q4-FT even slightly outperforms BanglaBERT, showing that decoder-only LLMs can match encoder-only BERTs when finetuned. We additionally include results from Faria et al. (2024), where they find GPT-3.5 with 15-shot examples (Brown et al., 2020) significantly outperforms even finetuned models. We note that Faria et al. (2024) only tested 300 random samples of XNLI-bn (out of 4895) due to the high cost associated with few-shot prompting.

4.6 Reward Modeling

As prefaced in Section 2.1, we created a Reward Modeling task where we asked LLMs to choose the better summary of an article. See Appendix B for the exact instruction used.

Table 7 shows that LLaMA-3-8B largely fails to pick the correct summary, be it in English or Bengali. LLaMA-3-70B and GPT4o are evenly matched on the English dataset while LLaMA-3-8B performs close to random chance (50%). We manually inspect LLaMA-3-8B’s outputs and find that it prefers the verbatim nature of using the first sentence as the summary (Example LLaMA-3-8B output: “Summary 2 is better because it aligns closely to the article and

Dataset	Model	Acc.
XLSum-en-300	LLaMA-3-8B	58.67
	LLaMA-3-70B	87.33
	GPT4o	87.33
XLSum-bn-300	LLaMA-3-8B	53.67
	LLaMA-3-70B	67.67
	GPT4o	63.33
translated-XLSum-bn-300	LLaMA-3-8B	65.33
	LLaMA-3-70B	73.33

Table 7: Bengali Reward Modeling performance of LLMs. translated-XLSum-bn-300 denotes XLSum-bn-300 translated into English using NLLB-1.3-Distilled.

does not include speculation or sensationalism.”). In Bengali articles, the performance of both LLaMA-3-70B and GPT4o degrade substantially.

Since the output of reward models are usually language-agnostic, numeric, or binary values, we explore whether translating the Bengali article and summaries using an automated translator can recover the lost performance. Specifically, we translate the Bengali articles to English using NLLB-1.3B-Distilled (Costa-jussà et al., 2022) and reattempt Reward Modeling on the translated dataset. This marginally recovers the accuracy of LLaMA-3-70B from 67.67% to 73.33%. However, assuming humans have a 100% accuracy on this task [§], this wide gap between human and LLM preference bodes ill for using English LLMs as reward models for Bengali LLMs.

5 Discussion

In Section 5.1, we discuss potential issues with existing Bengali downstream tasks. In Sections 5.2 and 5.3, we present key arguments for and against training a Bengali LLM in the short term.

5.1 Pitfalls of Machine-Translated Datasets

Table 1 shows that 3 out of 8 datasets we used were machine-translated. Machine translation is a cost-effective alternative to manual data annotation that requires much less human labor Li et al. (2023). However, this risks translation errors being propagated through translated datasets, leading to second-order effects on LLM training and evaluation. Even if there are no errors, stylistic

[§]A reasonable assumption since reference summaries were written by professional BBC contributors and the alternate summary is the article’s first line.

choices by automated translators can bias the dataset, something that is mitigated when there are multiple human annotators with different styles. We highlight such a case in Section 4.3 on the BanglaParaphrase (Akil et al., 2022) dataset.

5.2 A Case for Training Bengali LLMs

Better Generalization: Our experiments show that English-only LLMs surpass fine-tuned BanglaT5 on NLU tasks while performing well in NLG datasets. Furthermore, Asai et al. (2023) shows that well-known emergent capabilities of monolingual LLMs such as Instruction Tuning (Wei et al., 2021) and In-Context Learning (Brown et al., 2020) are less pronounced in other languages.

Better Tokenization and Efficiency: Yuan et al. (2023) shows that Bengali falls within the category of *Stagnant Languages*, i.e. does not noticeably improve if finetuned. The authors suspect this stagnation against finetuning occurs in languages, including Bengali, that are tokenized excessively and therefore are information-sparse. Excessive tokenization is also harmful from a performance perspective due to the $\mathcal{O}(n^2)$ time complexity and $\mathcal{O}(n)$ memory requirements of the standard attention mechanism in transformer-based LLMs.

Success of Chinese-oriented LLMs: The rapid progress of Chinese and English-Chinese bilingual LLMs (Baichuan, 2023; Cai et al., 2024; DeepSeek-AI, 2024; Bai et al., 2023) are particularly inspiring. Larger skews of Chinese LLMs such as Qwen-2-72B (Bai et al., 2023) and DeepSeek-V2-236B-MoE (DeepSeek-AI, 2024) far outperform GPT4 (Achiam et al., 2023) (90.1 vs. 70.95) on Chinese-MMLU (Li et al., 2023). Even smaller variants such as Baichuan2-13B (Baichuan, 2023), InternLM2-7B, and -20B (Cai et al., 2024) exhibit strong bilingual ICL capabilities.

The promise of a more capable and efficient Bengali Natural Language Generation, coupled with the proven success of Chinese LLMs are strong reasons to build a Bengali or English-Bengali LLMs. In fact, there have already been nascent attempts at such in the form of BanglaGPT (Salim et al., 2023), a GPT2-1.5B-based Bengali-only LLM.

5.3 A Case Against Training Bengali LLMs

Training Costs: Although exact training costs have not been released, it is rumored that LLaMA-3-8B cost Meta around 5 million USD on energy costs alone (Karpathy, 2024). Meta built

two custom 24K GPU superclusters[¶] and trained LLaMA-3-8B $\times 75$ longer than the Chinchilla (Hoffmann et al., 2022) optimal point. Even more efficient architectures and training recipes such as JetMoE-8B (Shen et al., 2024) required about 100K USD to train and it performs significantly worse than LLaMA-3-8B.

Limited Bengali Data: Beyond sheer training costs, the lack of high-quality Bengali datasets is another significant constraint. Currently, the largest Bengali pretraining corpus Bhattacharjee et al. (2021) is around 30GB while the largest open-source English corpus, FineWeb Penedo et al. (2024) is 36.7 TB. Bengali also lacks the necessary RLHF datasets for instruction-tuning LLMs, a crucial step that aligns LLMs to human preferences and values.

The training of smaller LLMs such as LLaMA-3 (Abdin et al., 2024) or the Phi series (Abdin et al., 2024) is highly iterative and heavily dependent on being able to filter out low-quality data with older LLMs and generating high-quality synthetic (*textbook quality*) data with larger LLMs such as GPT4 (Abdin et al., 2024). Limited training data and the lack of preexisting Bengali LLMs create a negative feedback loop when attempting to train LLMs for Bengali.

Rapid Progress of Closed-source LLMs: Any attempt to train a large-scale Bengali-oriented LLM may be premature due to the possibility of frontier AI labs increasing support for Bengali. For example, the latest model by OpenAI, GPT4o, reduced the token count of non-Latin scripts by as much as 4.4 times compared to GPT4-Turbo^{||}. Better Bengali support in frontier LLMs would significantly help synthetic data generation.

Building two-staged pipelines with state-of-the-art translation (Costa-jussà et al., 2022) and English LLMs might be a better research direction in the short term while also being a significant stepping stone towards training LLMs for Bengali.

5.4 Next Step towards Bengali LLMs: Compiling Pretraining Corpus

Both BanglaT5 (Bhattacharjee et al., 2022) and BanglaBERT (Bhattacharjee et al., 2021) were trained on a new pretraining corpus Bangla2B+, collected by the authors via web-crawl on 110 Ben-

gali websites. Alternative data sources include OSCAR (Abadji et al., 2022) and CCNet (Wenzek et al., 2020) but they reportedly contain noise and offensive text that is infeasible to filter out.

Bangla2B+ totals about 35 GB of data, a far cry from English pretraining corpora of 44 TB. Expanding the selection of scraped websites, transcribing Bengali media content, digitizing printed media, and translating English corpora using automated translation pipelines are possible methods to increase the Bengali pretraining corpus. As we have highlighted in 4.3, naive translation induces bias. We advocate for using multiple translation models in tandem, advanced prompting, and rigorous post-processing to ensure not just accuracy, but also linguistic naturalness and consistency.

6 Other Related Works

Alongside models pretrained exclusively on Bengali script such as BanglaT5 (Bhattacharjee et al., 2021) and BanglaBERT (Bhattacharjee et al., 2022), there exists multilingual models trained on related Indic languages including Bengali. These include encoder-only transformers such as MuRIL-BERT (Khanuja et al., 2021), IndicBERT (Dodapaneni et al., 2022), and encoder-decoder transformers such as IndicBART (Dabre et al., 2021).

Besides the datasets in Table 1, other Bengali downstream tasks include grammatical error detection and correction, sentiment analysis, and transliteration. Oshin et al. (2023) introduced a new Bengali error detection dataset and correction and found that BanglaBERT (Bhattacharjee et al., 2021) excels at detection while BanglaT5 (Bhattacharjee et al., 2022) excels at correction. However, Shahgir and Sayeed (2023) finds that BanglaT5 can match BanglaBERT on detection, albeit on a different dataset (Md Boktiar Mahbub Murad, 2023). Elahi et al. (2024) finds that BanglaBERT outperforms MuRIL (Khanuja et al., 2021) on both noisy and noise-reduced Bengali sentiment analysis (Islam et al., 2021). Roark et al. (2020) introduces a Bengali transliteration dataset and finds that a transformer (Chen et al., 2018) outperforms LSTMs at the task.

Asai et al. (2023) compares downstream task performance in multiple languages including Bengali. The authors find that in-context learning with LLMs such as BLOOMZ-7B, BLOOM-176B (Workshop et al., 2022) and GPT-3.5 underperform compared against fine-tuned mT5 (Muennighoff

[¶]<https://ai.meta.com/blog/meta-llama-3/>

^{||}<https://openai.com/index/hello-gpt-4o/>

et al., 2022) baselines. Similarly, a concurrent work Kabir et al. (2023) finds that fine-tuned BanglaT5 and BanglaBERT outperforms GPT-3.5, LLaMA-2-7B (Touvron et al., 2023) and Claude 2 **. In contrast, we test more recent and capable LLMs including LLaMA-3 (Meta, 2024), GPT4 Achiam et al. (2023) and find that LLMs outperform fine-tuned models in multiple Bengali benchmarks.

7 Conclusion

Our experiments reveal a mixed landscape; while LLMs generally outperform fine-tuned T5 baselines on Bengali NLU tasks, their performance on Bengali NLG tasks, particularly those requiring Bengali script generation, leaves room for improvement. Key findings include the inefficient tokenization of Bengali script by existing LLMs, task-dependent performance variations, and potential biases in machine-translated datasets. The study also highlights the significant costs and data requirements for training Bengali-specific LLMs, balanced against the rapid progress in the multilingual capabilities of existing models. In the short term, leveraging state-of-the-art translation models with powerful English LLMs may offer a pragmatic approach to improving Bengali language technologies. Future research should focus on developing more efficient tokenization methods for non-Latin scripts, creating high-quality Bengali datasets, and utilizing cross-lingual transfer.

8 Limitations

Lack of Human Evaluation : While we identified the need for human evaluation in tasks like paraphrasing, we did not conduct human evaluations ourselves due to resource constraints. This limits our ability to fully assess the quality of model outputs, especially for generation tasks.

Tokenization Analysis : Although we identified inefficiencies in Bengali script tokenization, a more in-depth analysis of its impact on model performance across different tasks and model sizes could provide further insights.

Fine-tuning Experiments : The evaluation of larger models was limited by available computational resources. Our fine-tuning experiments were limited in scope and primarily focused on LLaMA-3-8B. A more comprehensive exploration

of fine-tuning across different model architectures and sizes could yield additional insights.

Temporal Limitations : Given the rapid pace of development in the field of LLMs, some of our findings may become outdated as new models and techniques are introduced.

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**<https://www.anthropic.com/news/claude-2>

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A Finetuning Hyperparameters

We finetune LLaMA-3-8B-Instruct using QLoRA ($r = 16$, $\alpha = 16$) and 4-bit integer quantization with learning rate 5×10^{-5} , warmup-ratio 0.05 and linear-rate scheduling for all tasks. We use a single Nvidia RTX A6000 for all experiments.

On BanglaParaphrase (Akil et al., 2022), we train for 1 epoch with batch size 32 and gradient accumulation every 4 batches.

For BanglaRQA (Ekram et al., 2022), we train for 3 epochs with batch size 4 and gradient accumulation every 8 batches. We filtered out training and validation samples with context smaller than 500 and longer than 3900 characters for efficient training.

For XNLI_bn (Bhattacharjee et al., 2021), we train for 1 epoch with batch size 32 and gradient accumulation every 4 batches. We filtered out training samples where the combined length of the two sentences was less than 50 and longer than 350 characters for efficient training.

We did not extensively tune hyperparameters for any fine-tuning experiments.

B Prompts

Translation

SYSTEM:

You are a state-of-the-art AI assistant that translates sentences from {Language A} to {Language B}. The user provides you with a {Language A} sentence, and your task is to translate it into {Language B}. Just return the translation without any preamble, quotations, or explanations.

USER: {Language A sentence}

Paraphrase Generation

SYSTEM:

You are a state-of-the-art AI assistant that generates Bengali paraphrases. The user provides you with a Bengali sentence, and your task is to generate a Bengali paraphrase of it. Just return the paraphrase without any preamble, quotations, or explanations.

USER: {Bengali sentence}

Summarization

SYSTEM:

Please write a one-sentence {Language A} summary/TL;DR of the given {Language B} article. The summary must be in {Language A} and not be longer than a sentence. Just return the summary without any preamble, quotations, or explanations.

USER: {Language B Passage}

Natural Language Inference

SYSTEM:
You will be given two sentences. Please determine whether the first sentence entails, contradicts, or is neutral to the second. Pay close attention to each word as you analyze the relation between the two sentences. Respond in the format:
Thought: {thought on if the first second entails, contradicts, or is neutral to the second sentence}\n\nVerdict: {any one of <entailment>, <contradiction> or <neutral> tags}

USER:
Sentence 1: {}\n\nSentence 2: {}

BQA/Squad-bn

SYSTEM:
Is the to the question in the context? ('YES'/'NO'). What is the answer? (A substring of the context/'<NOT_IN_CONTEXT>'). Return as a tuple (eg. ('YES', answer_substring) or ('NO', '<NOT_IN_CONTEXT>') without any preamble or explanations.

USER: Context: {context} \n\n Question: {question}

BanglaRQA

SYSTEM:
The user will provide a context and a question, both in Bengali.
Read the context and the question carefully.

Respond with a JSON object with the following keys:

"answerable" (boolean, Is the question answerable from the context?)
"question_type" (yes-no / single-span / multiple-span)
"answer" ('Yes' or 'No' for yes-no questions)/substring of the context for single-span/list of substrings of the multiple-span/'<NOT_IN_CONTEXT>')

USER:
Context: {}\n\nQuestion: {}

We used the Bengali words for 'Yes' and 'No' when specifying the *answer* key in the above prompt.

Reward Modeling

UESR:

Here is a news article:

```
<article>
{article}
</article>
```

Here is one person's summary of the article:

```
<summary1>
{summary1}
</summary1>
```

And here is a second person's summary of the same article:

```
<summary2>
{summary2}
</summary2>
```

Please read the article and both summaries carefully. Then, in `<thoughts>` tags, analyze the strengths and weaknesses of the two summaries, focusing on the following criteria:

- 1) Faithfulness - does the summary accurately reflect the key points of the article without adding extraneous or false information?
- 2) Coherence - is the summary well-structured and easy to understand?
- 3) Concision - does the summary capture the essence of the article efficiently, without unnecessary detail or repetition?
- 4) Abstraction - does the summary rephrase the article content in novel ways, or does it just extract verbatim snippets?

Favor summaries that demonstrate abstraction and rephrase content in their own words over ones that just extract verbatim snippets.

After you've thought it through, provide your final verdict on which summary is better inside `<verdict>` `</verdict>` tags, using either a `<first>` or `<second>` tag to indicate your choice. You must pick one or the other, you cannot hedge or say they are equal. The summary that does a better job meeting the above criteria, especially abstraction, should be selected as the better one.

BEnQA

SYSTEM:

You are given a multiple choice question and their options in English/Bengali and your job is to correctly answer the question. First reason step by step in English/Bengali and only then give me the final answer as "a", "b", "c" or "d".

Keep these in mind:

1. Only include the letter a, b, c, d as your final answer. Do not include the option text.
2. Every question will have an answer in the given options. So, DO NOT say that none of the answers are correct.
3. ONLY ONE of the given options will have the answer. So DO NOT provide multiple options as answers.
4. The questions contain enough information to solve the problem, so DO NOT say that you need additional information.
5. Answer in the format:
\n'Let's think step by step.\n{reasoning}\n\nAnswer:{A/B/C/D}'

USER:

Question:

{Bengali question}

Options:

{Bengali options}

C Tokenization of Bengali Script by English-oriented Language Models

Tokenizer	$ Context $	$ Vocab $	English	Bengali
BanglaT5	512	32K	3.05	5.09
NLLB	1024	256K	4.25	3.35
AYA-23	8192	255K	4.75	0.87
LLaMA-3	8196	128K	4.77	0.83
Mistral	32768	32K	4.31	0.90
Qwen2	131072	152K	4.69	0.94

Table 8: Average character per token values of different tokenizers on 11535 English and 8012 Bengali BBC articles from XLSum (Hasan et al., 2021a).

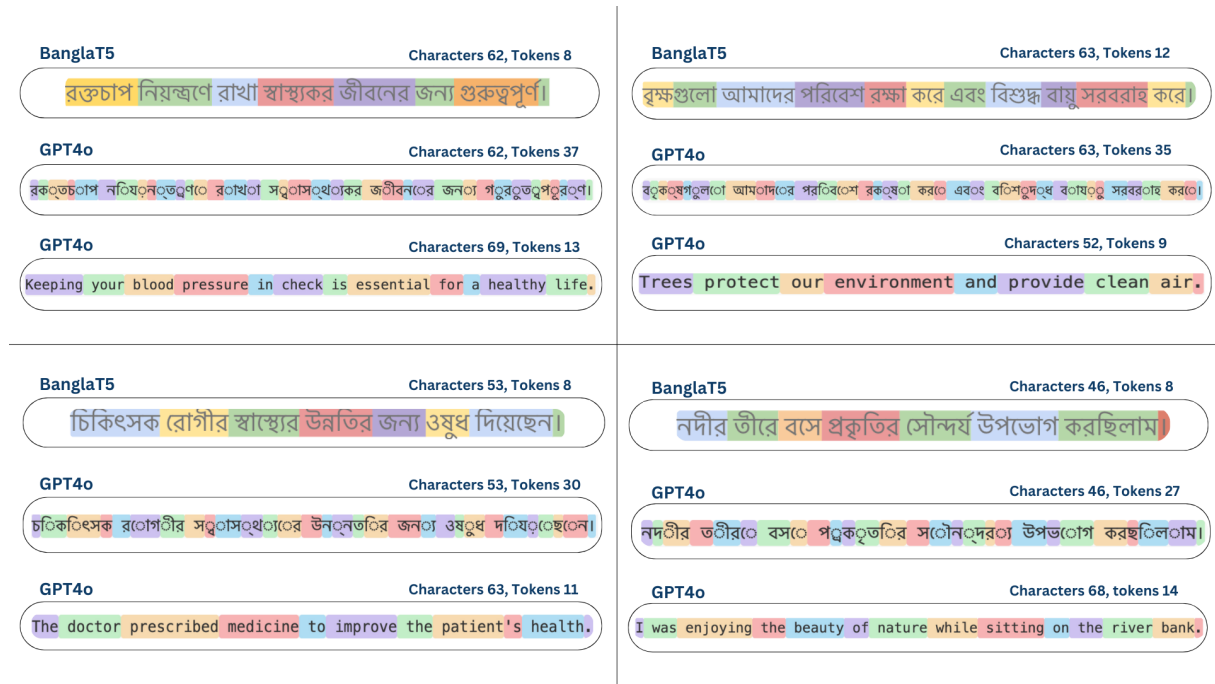


Figure 2: Qualitative Examples of Inefficient Bengali Script Tokenization.

D Additional Results

D.1 BEnQA

Subject	Total	LLaMA-3		GPT3.5	GPT4
		8B	70B		
8th-Math	209	0.584	0.722	0.486	0.808
8th-Science	228	0.465	0.640	0.356	0.721
10th-Biology	351	0.499	0.638	0.351	0.775
10th-Chemistry	389	0.494	0.658	0.404	0.741
10th-Math	380	0.453	0.700	0.407	0.775
10th-Math-II	393	0.478	0.695	0.383	0.781
10th-Physics	319	0.47	0.639	0.36	0.75
12th-Biology-I	310	0.445	0.603	0.346	0.721
12th-Biology-II	328	0.415	0.598	0.315	0.712
12th-Chemistry-I	367	0.469	0.638	0.314	0.775
12th-Chemistry-II	389	0.393	0.640	0.355	0.751
12th-Math-I	396	0.467	0.684	0.431	0.756
12th-Math-II	391	0.394	0.542	0.391	0.662
12th-Physics-I	304	0.457	0.664	0.375	0.774
12th-Physics-II	333	0.429	0.670	0.319	0.775
12th-Chemistry-I	367	0.469	0.638	0.314	0.775
Total/Avg	5087	0.457	0.648	0.372	0.751

Table 9: Subject-wise Accuracy in English.

D.2 NLLB

Model	Arch.	Parameters	E-B	B-E
NLLB-200	MoE	54.5B	50.0	62.2
NLLB-200	Dense	3.3B	48.7	61.1
NLLB-200	Dense	1.3B	47.3	59.8
NLLB-200-Distilled	Dense	1.3B	47.8	60.1
NLLB-200-Distilled	Dense	600M	46.2	57.9

Table 10: Translation Metric of the current state-of-the-art NLLB model family on the NLLB dataset (Costa-jussà et al., 2022). Reporting chrF++ scores.